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Time resolution limits of a fast SiPM

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Abstract: The Silicon Photomultiplier (SiPM) is a highly sensitive photon detector, widely used in various fields such as medical imaging and high-energy physics, due to its single-photon detection capability, good timing performance and magnetic insensitivity. The SiPM with Epitaxial Quenching Resistor (EQR) is an emerging technology that has recently captured the interest of the researchers.The growing demand for physics applications and the enhanced measurement precision of certain imaging devices have led to elevated expectations regarding the time performance of the SiPM. In this study, the limit time resolution of two EQR20 11-3030-S SiPMs produced by Novel Device Laboratory (NDL) was evaluated using a femtosecond laser. The time resolution limits of the two samples result in 2.25 ps and 2.88 ps.

Keywords: Photon detectors for UV, visible and IR photons (solid-state); Timing detectors

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1 Introduction

The Silicon Photomultiplier (SiPM) is a highly sensitive solid state photodetector made of an array of hundreds or thousands of integrated Single-Photon Avalanche Diodes (SPADs), called microcells or pixels [1]. The SiPM is characterized by its compact size, immunity to magnetic fields, low power consumption, and good single-photon resolution capability. Therefore, it is widely used in various fields, such as medical equipments [2, 3], high energy physics [4, 5], etc. To broaden the application of SiPMs in various fields such as TOF-PET, an assessment of their timing performance is essential. For SiPM, the Single-Photon Time Resolution (SPTR) is commonly employed to characterize the timing performance and has been extensively investigated in previous studies [6]. The timing resolution limit is also a useful parameter to evaluate the time performance of SiPM, for example, the time resolution limit of SenSL-J30035 SiPM was measured to be 2.68 ps [7].

The epitaxial quenching resistor (EQR) SiPM [8] developed by Novel Device Laboratory (NDL) employs epitaxial substrate resistor instead of polysilicon resistor to connect avalanche photodiode (APD) cells. As alternative detectors for CEPC-ECAL [9], several performance characteristics of various EQR SiPMs have been evaluated [8]. However, there has been limited research on their timing performance. In this study, a femtosecond laser was used as a light source to calibrate the intrinsic time resolution limit of the EQR20 11-3030-S SiPM, with an effective pitch of 20 µm and an active area of $3.00 \times 3.00\,\mathrm{mm}^2$.

2 Experiment setups

In traditional photon detector Transit Time Spread (TTS) measurements, the synchronization signal provided by a laser generator is typically employed as the trigger. However, inherent uncertainties in this signal can lead to inaccurate results for ultrafast timing performance measurements. Basing on previous research [7], we utilized a femtosecond laser generator (Spitfire ACE-1001HP from Spectra-Physics) as the light source, with its emission peak adjusted to 400 nm, operating at a frequency of 1 kHz, and a pulse width less than 120 fs. As shown in figure 1, to reduce the light intensity, the intense light generated by the laser passed through two neutral optical attenuators before entering the fiber optic system. A spliced one-to-two quartz fiber splitter was employed to ensure that the light beam reached both detectors simultaneously, thereby minimizing uncertainties caused by the light source system. The fibers are connected to the cathode of the detectors. The time jitter between the signals of the two detectors is used to evaluate the intrinsic time resolution limit of the detectors.

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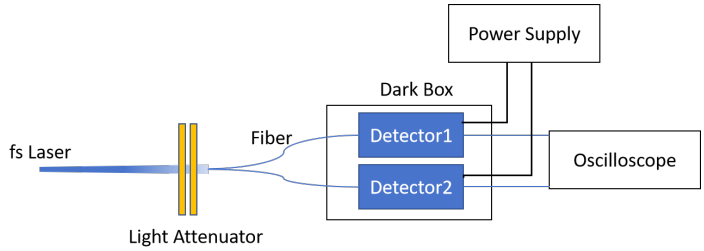

Figure 1. Setup for time resolution measurements using a femtosecond laser.

To reduce measurement uncertainties, the Photek 210 PMT with fast timing performance was used as the trigger signal. To efficiently capture the ultra-fast waveforms produced by the PMTs, a LeCroy HDO9404 oscilloscope with a bandwidth of 4 GHz and a sampling rate of 40 GS/s was employed as the data acquisition system. The waveforms were recorded by the high-performance oscilloscope for offline analysis and processing.

3 Result and discussion

First, responses of two identical single-anode Photek 210 MCP-PMTs to light pulses from the femtosecond laser were tested. Table 1 presents performances of the two PMTs under normal condition, including operating High Voltage (HV), gain, Rise Time (RT), Fall Time (FT), Pulse Width (PW), and TTS at Single Photon Electron (SPE) mode. To prevent PMTs damage caused by intense laser light pulses, the gains of both PMTs were adjusted to be less than 10^3. By adjusting the voltage,

Table 1. Parameters of two Photek 210 PMTs under normal condition.

PMT	HV	Gain	RT	FT	PW	TTS@SPE
PMT1	−4300 V	8.5×10^5	132.8 ps	318.9 ps	207.3 ps	31.7 ps
PMT2	−5000 V	1.1×10^6	114.1 ps	114.9 ps	162.6 ps	31.2 ps

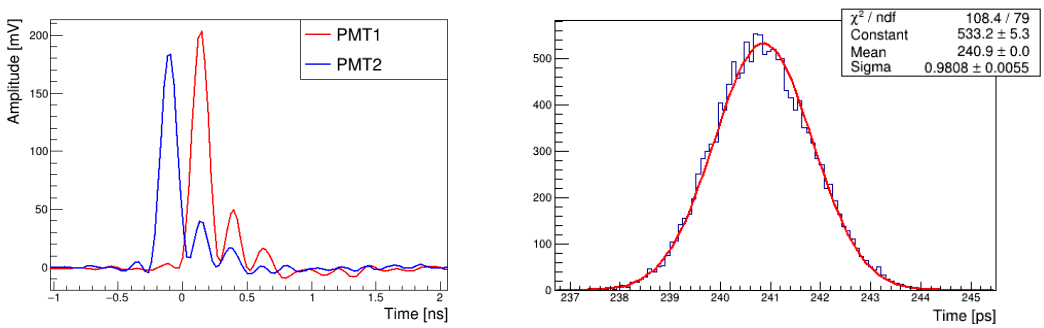

Figure 2. Typical waveforms of PMTs (left) and the distribution of time difference between the two PMTs signals (right).

the gain levels of the two PMTs were made consistent. As shown in figure 2, at the same light intensity, the signals from the two PMTs were similar in amplitude. Using lead edge discrimination, with a 50% pickup threshold applied, the time resolution was measured. The time jitter σ_{measured}

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between the two detectors can be described as:

$$\sigma_{\text{measured}} = \sqrt{\sigma^2_{\text{Detector1}} + \sigma^2_{\text{Detector2}} + \sigma^2_{\text{system}}} \tag{3.1}$$

The σ_{measured} between the two PMTs signals was 0.98 ps. Since the two PMTs are of the same model, we can assume that the two PMTs have consistent time resolution at the same gain. The time jitter of channels of the oscilloscope is 0.5 ps [10]. Therefore, the time resolution σ_{PMT} is equal to $\frac{\sqrt{\sigma^2_{\text{measured}} - \sigma^2_{\text{system}}}}{\sqrt{2}}$, which is 0.60 ps.

Table 2. Parameters of two SiPMs under normal condition.

SiPM	Voltage	Gain	RT	FT	Pulse Width	TTS@MPE
SiPM1	29 V	8.2×10^5	0.33 ns	7.32 ns	2.92 ns	11.1 ps
SiPM2	29 V	8.2×10^5	0.35 ns	7.35 ns	2.92 ps	12.6 ps

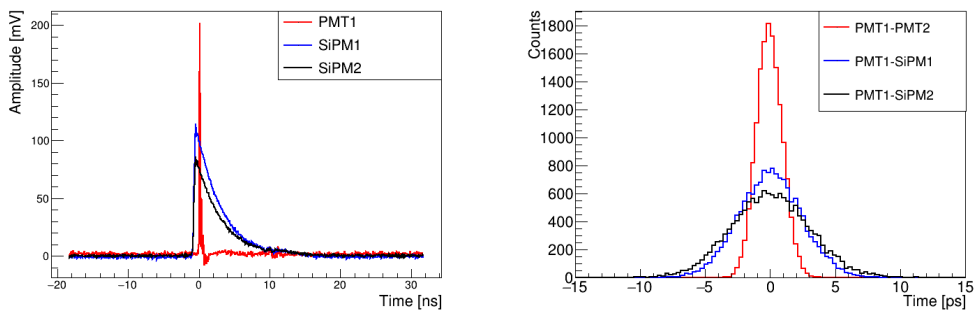

Figure 3. Typical waveforms of two SiPMs and a PMT (left) and their time distributions (right).

Table 2 presents the performance of the two SiPMs under normal condition. One of Photek 210 PMT was used as the trigger for the other detectors. Figure 3 shows the typical waveforms of the two SiPMs (SiPM1 and SiPM2). The σ_{measured} can be described as:

$$\sigma_{\text{measured}} = \sqrt{\sigma^2_{\text{PMT}} + \sigma^2_{\text{SiPM}}} \tag{3.2}$$

As shown in figure 3 right, the standard deviation of the distribution denotes the time resolution limits of SiPM1 and SiPM2 are 2.39 ps and 2.99 ps respectively. Subtracting the time jitter of the trigger signal, the intrinsic time resolution of SiPM1 and SiPM2 are 2.25 ps and 2. 88 ps respectively.

4 Conclusion

In this manuscript, the time resolution limits of two ERQ SiPMs developed by NDL were measured using a femtosecond laser. The Photek 210 PMT with ultrafast time performance was utilized for generating trigger signals, and a high-sampling-rate oscilloscope was employed to record the signals waveforms. The time resolution limits of two samples of NDL EQR20 11-3030-S SiPMs were measured. The limits for samples SiPM1 and SiPM2 are 2.25 ps and 2.88 ps respectively. These findings extend the applicability of SiPMs in applications requiring high time resolution, demonstrating their potential in academic research and scientific applications.

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